

Pouya Kia

Data Engineer

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Summary

Data Engineer with 5+ years of experience building and maintaining production data pipelines in fintech and payment environments. Strong in Python, SQL, Airflow, data modeling, and batch and streaming ETL, with a focus on data quality, performance, and reliability. Experienced in building scalable data solutions and working with large transactional datasets across different systems. Comfortable with AWS cloud services and familiar with Terraform and infrastructure as code, with hands-on cloud experience primarily on GCP.

Skills

Programming Languages: Python, SQL

Version Control & DevOps: Git & GitHub, Docker & Kubernetes, Jenkins, CI/CD

Cloud Platforms: AWS (Familiar), GCP, BigQuery

Database Technologies: PostgreSQL, MySQL, MongoDB, Redis

ETL & Data Pipelines: Apache NiFi, Airflow & dbt, Talend, Luigi

Data Modeling & Architecture: Kimball, Data Vault, Snowflake Schema

Data Engineering & Tools: Apache Kafka, Apache Spark, Apache Airflow, dbt

Infrastructure as Code: Terraform (Familiar)

Soft Skills: Problem Solving, Communication, Collaboration

Work Experience

Bitex

Senior Data Engineer

Tehran, Iran

2025/02 - Present

An OTC online exchange for trading cryptocurrency.

- Built a real-time data ingestion pipeline for fraud detection, streaming transaction, login, and device data via Kafka and storing in a feature store for ML.
- Implemented a Docker-based development workflow for data pipelines, using Docker Compose for local development and deploying production workloads via Kubernetes, ensuring consistent environments and CI/CD integration for scalable, testable deployments.
- Designed and optimized ETL pipelines with Apache Airflow (CeleryExecutor), reducing lock contention on reporting tables by implementing a temp-table swap pattern, cutting Metabase query time significantly through PostgreSQL index tuning, query planner optimization, and server configuration tuning (shared_buffers, work_mem).
- Built incremental data sync pipelines from MariaDB to PostgreSQL using server-side streaming cursors, eliminating memory exhaustion on large tables and reducing source DB connection timeouts on multi-million row transfers.
- Led a data migration project from MongoDB to MariaDB, re-architecting the schema for performance and scale, improving query performance by 30% and supporting 50% more traffic.
- Implemented pipeline observability with automated failure alerting via Mattermost webhooks, firing only on final retry failure with localized timestamps, and automated infrastructure resilience with Docker health-check watchdogs for Airflow scheduler to prevent undetected outages.

Asan Pardakht

Data Engineer

Tehran, Iran

2023/06 - 2025/01

A leading payment service provider in Iran, processing millions of daily transactions across digital payment and wallet services.

- Containerized ETL and analytics workloads using Docker, deployed to Kubernetes clusters for cloud-native, reproducible execution, and integrated into CI/CD pipelines via GitHub Actions.
- Built and deployed scalable batch and streaming data pipelines handling over 2TB daily using Apache Kafka, Spark, and Airflow, leveraging real-time data processing on GCP Cloud Storage and Pub/Sub to improve throughput and efficiency.
- Developed and maintained 5+ data marts using a hybrid Kimball and Data Vault modeling strategy, orchestrated with Apache Airflow and dbt for modular, testable, and auditable data models.
- Standardized SQL schemas with lineage-aware table structures, implementing rigorous data validation, cutoff timestamps, and dbt tests to ensure end-to-end data integrity and observability.

Asan Pardakht

Data Scientist

- Developed a Next-Best-Action model and churn predictor (XGBoost, 92% accuracy), boosting engagement by 10% and reducing churn by 18%.
- Designed forecasting services for 50M+ users, utilizing RNNs to achieve over 90% precision, enhancing user experience and decision-making.
- Defined and tracked key performance indicators (KPIs) through analyzing a staggering volume of 10 million daily transactions, ensuring accurate representation and insightful analysis for informed decision-making.

Tehran, Iran

2023/01 - 2023/05

Otaghak

Data Scientist

An online travel and accommodation booking platform.

- Engineered a recommendation system using tree-based models and matrix factorization algorithms, increasing conversions by 65%.
- Deployed ML models to predict customer behavior, leading to a 19% increase in engagement.
- Conducted RFM-based segmentation to drive a 10% sales uplift and crafted targeted marketing strategies.
- Formulated and executed a Power BI dashboard, enhancing operational efficiency by 25%.

Tehran, Iran

2021/01 - 2022/01

Dayche Group

Junior Data Scientist

A consulting and project-based technology services company delivering data, analytics, and software-related projects for clients.

- Analyzed 10K+ lung images using OpenCV & Scikit-image, achieving 95% diagnostic accuracy in infection detection.
- Integrated an OCR pipeline with Python, accelerating data processing by 40%.
- Contributed to predictive modeling efforts that improved sales efficiency by 8%.

Tehran, Iran

2020/01 - 2020/12

Education

University of Tehran

Master of Science - Industrial Engineering (Optimize systems)

Tehran, Iran

2017 - 2021

Qom University of Technology

Bachelor of Science - Telecommunication Engineering

Qom, Iran

2012 - 2016

Languages

English (B2)